Crisil

a company of S&P Global

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https://crisil-financial-nlp.pages.dev

Market Pulse

S&P Global & Crisil Campus Hackathon

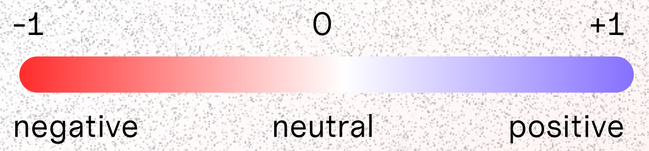
Objectives & Research Findings

We first researched about existing datasets and models relevant to our objectives and made recommendations

Our Objectives

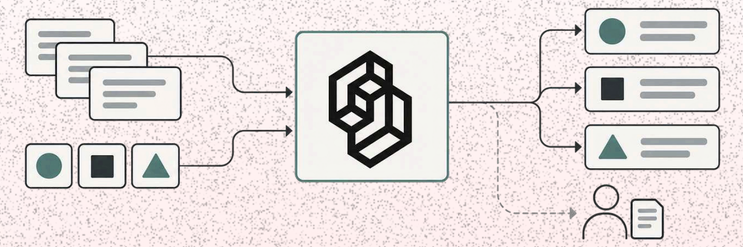
Sentiment Analysis

1. Encode financial news using pretrained text encoders such as **FinBERT** or **FLANG-RoBERTa**, trained on financial corpora.
2. **Available datasets:**
   - **FiQA:** Financial sentiment scores for regression.
   - **Financial PhraseBank:** Positive, negative, and neutral sentiment labels.
   - **Financial tweets and headlines:** Useful for sentiment classification on short-form text.
3. Fine-tune a classification head for sentiment labels or a regression head for continuous sentiment scores (e.g., −1 to +1).



Event Classification and Impact Scoring

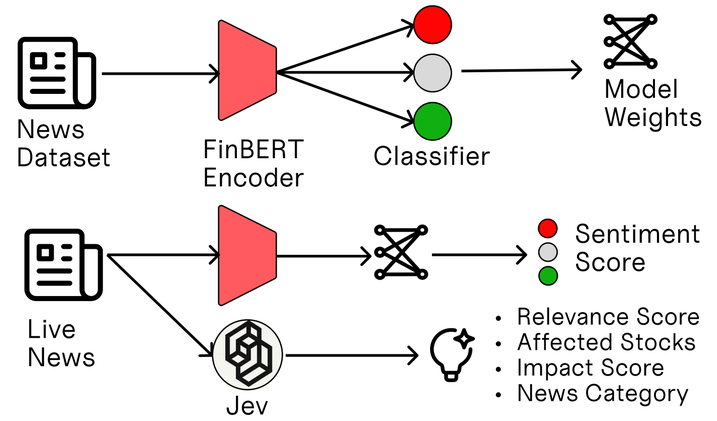
1. **Potential tasks include:**
   - Stock/ticker classification.
   - News relevance scoring.
   - Event classification.
   - Potential market-impact scoring (e.g., 1–10).
2. Use decision LLMs such as **Jev** or open-source models such as **Laya** for zero-shot tasks where labelled datasets are unavailable.
3. **Decision-making LLMs** are specialized AI systems designed to output direct classifications, scores, or choices instead of long-form text.



Structured Trade and Risk Signals

1. **Keyword matching:** Match company names, ticker symbols, and aliases against news text.
2. **Embedding similarity:** Compare news encodings with stock/company descriptions to identify potentially affected stocks.

Plausible Approach



Experiments and Results

We experimented with different models and finetuned and tested them on benchmarks to select the most appropriate models for sentiment scoring, affected stocks classification, relevance and impact scoring etc.

Sentiment Classification (-1,0,1) :
Compares three financial-domain BERT models using average macro-F1 and accuracy across Financial PhraseBank and Twitter Financial News.

Model	Average F1	Average Accuracy
FLANG-RoBERTa	**85.51%**	**87.36%**
FLANG-DistilBERT	82.51%	85.17%
FinBERT	81.09%	84.27%

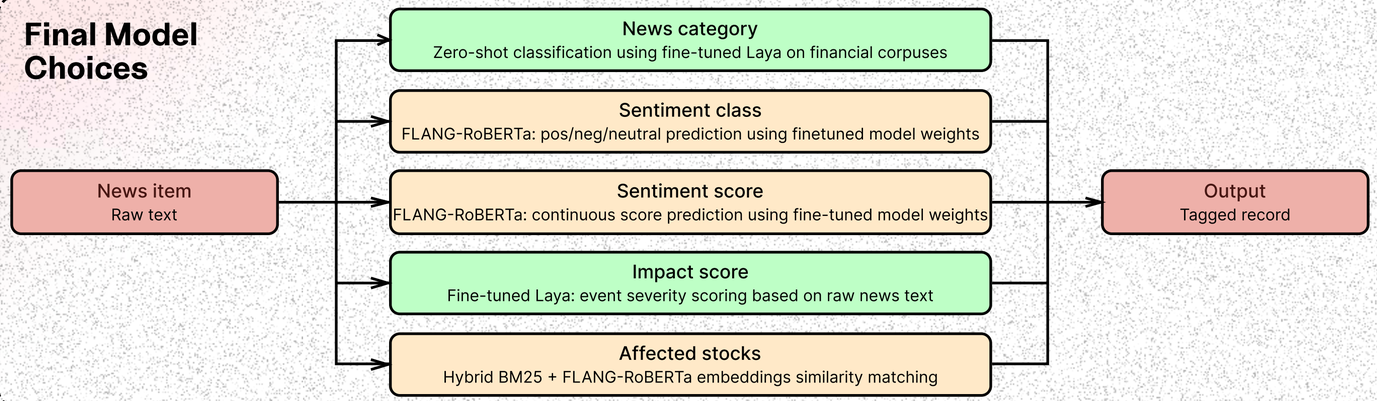
Sentiment Regression (-1 to 1) :
Evaluates the models' ability to predict continuous financial sentiment scores on FiQA using MAE and R^2.

Model	MAE ↓	R^2 Score ↑
FLANG-RoBERTa	**0.1848**	**0.4617**
FLANG-DistilBERT	0.1984	0.3948
FinBERT	0.2064	0.3071

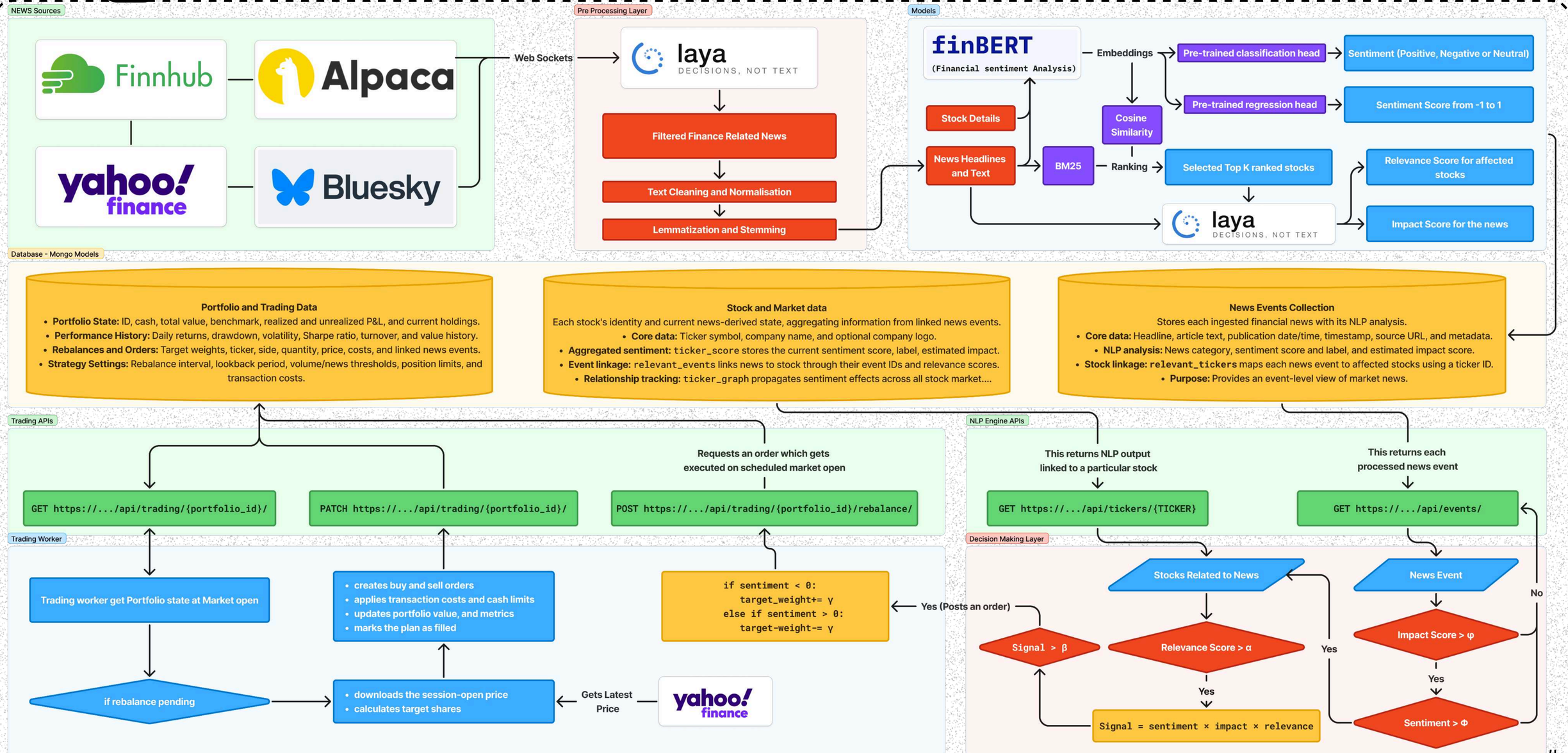
Laya vs Jev :
Compares base and fine-tuned Laya with zero-shot Jev across sentiment classification, sentiment regression, and ticker identification. Laya is an open-source light weight decision LLM which can be finetuned on datasets to improve domain specific results.

Model	Regression R^2	Average Classification F1	Ticker Classification Micro-F1
Laya Base	-2.0921	62.78%	6.86%
Laya Fine-tuned	0.3596	**84.85%**	**55.81%**
Jev (Zero-shot)	**0.4854**	76.78%	50.49%

Final Model Choices

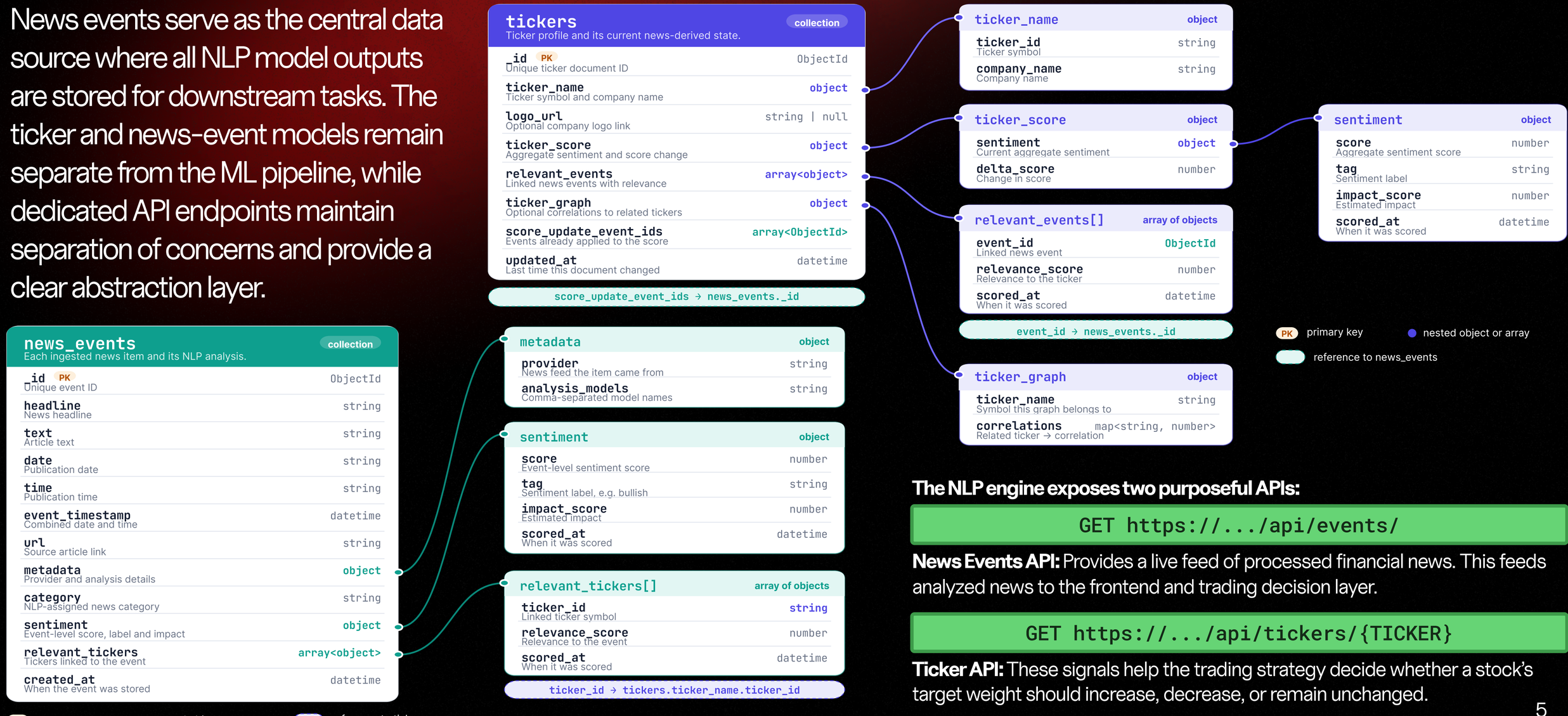


High Level Arhitecture



Data Modelling and LLD

News events serve as the central data source where all NLP model outputs are stored for downstream tasks. The ticker and news-event models remain separate from the ML pipeline, while dedicated API endpoints maintain separation of concerns and provide a clear abstraction layer.



The NLP engine exposes two purposeful APIs:

`GET https://.../api/events/`

News Events API: Provides a live feed of processed financial news. This feeds analyzed news to the frontend and trading decision layer.

`GET https://.../api/tickers/{TICKER}`

Ticker API: These signals help the trading strategy decide whether a stock's target weight should increase, decrease, or remain unchanged.

Trading Application and Results

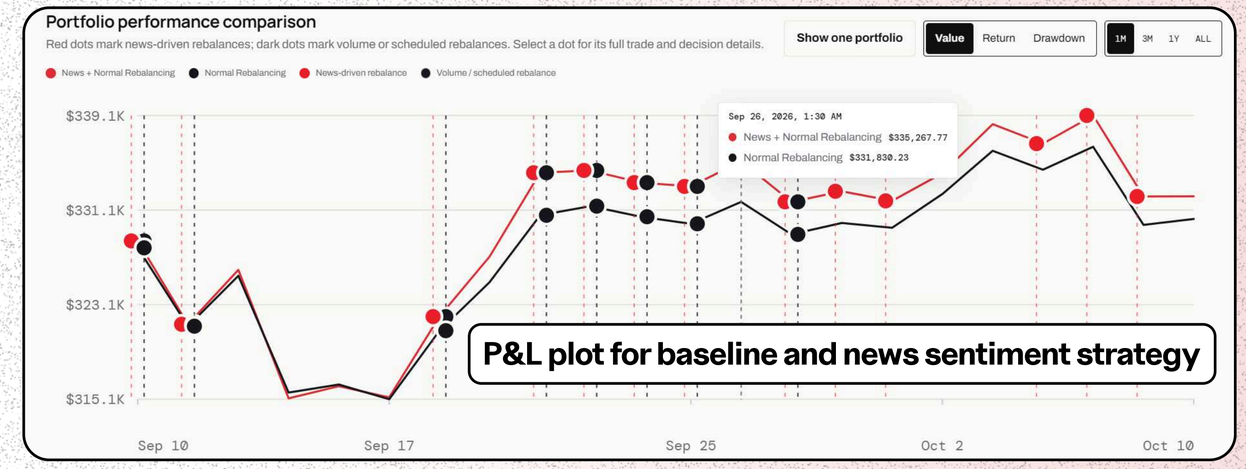
We build a baseline equity-trading strategy and add news sentiment as an additional signal to evaluate its impact on portfolio performance.

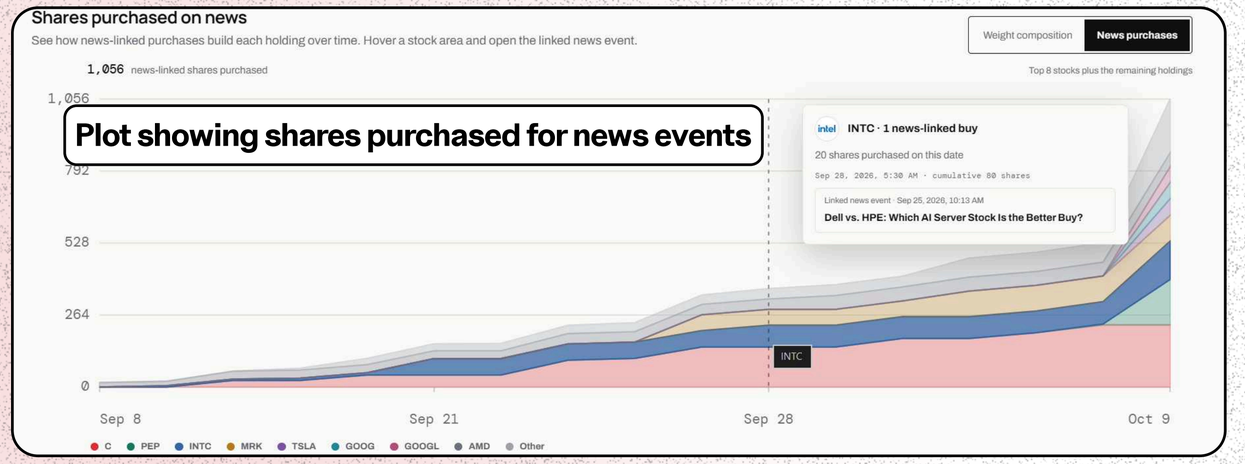
Metric (1 month)	Normal Rebalancing	News + Normal Rebalancing
Starting portfolio value	$324,350.44	$324,350.44
Final portfolio value	$332,159.69	**$338,408.51**
Profit / Loss	$9,809.25	**$14,058.07**
Return	2.41%	**4.33%**

NLP-Engine-Based Trading

1. **NLP Analysis:** Use FLANG models to assess sentiment, impact, & news categories.
2. **Stock Identification:** Identify affected stocks using semantic similarity and BM25.
3. **Signal Generation:** Combine sentiment, impact, and stock relevance into signals.
4. **Portfolio Rebalancing:** Adjust target weights subject to relevance thresholds.
5. **Trade Execution:** Execute approved orders at the next eligible market session, accounting for cash and transaction costs.

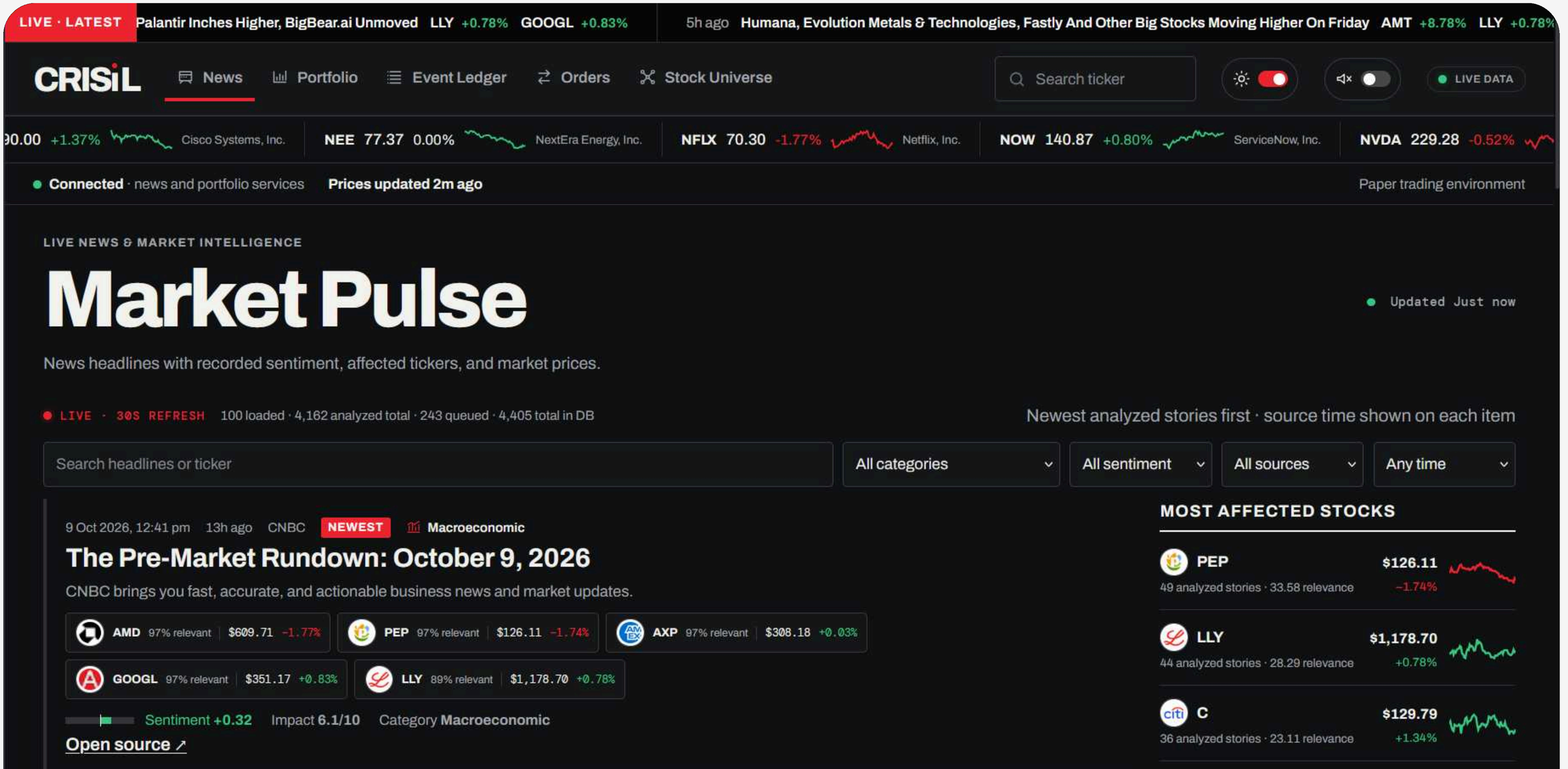
- **Outperformance:** News Sentiment enabled strategy delivered 0.69% higher return.
- **Trading activity:** 500+ news events were analyzed for 1 month, generating 39 ticker-level order rows across 14 execution dates, linked to 50+ distinct events.
- **Event-driven decisions:** Sentiment, impact, stock relevance, and correlation signals guide position adjustments alongside the existing rebalancing strategy.
- **Conclusion:** News signals improved returns in this historical replay.





User Interface

https://crisil-financial-nlp.pages.dev



Impact and Future Scope

The NLP engine converts large volumes of unstructured financial news into structured sentiment, impact, and stock-relevance signals. This helps investment teams identify market-moving information faster, reduce manual research, and connect news directly to affected companies. Its API-based design makes the analyzed outputs reusable for dashboards, screening tools, alerts, portfolio systems, and other financial applications.

Impacts

Centralized data layer: Processed news and ticker signals are stored centrally and exposed through APIs for the frontend, trading engine, charts, and future analytics.

Actionable trading signals: Sentiment and impact scores are converted into stock-level signals that can adjust portfolio target weights.

Explainable decisions: Every news-driven trade can be linked back to the analyzed headline, relevant ticker, sentiment score, impact score, and resulting rebalance plan.

Faster financial analysis: The NLP engine processes large volumes of news and extracts sentiment, impact, category, and relevant companies automatically.

Limitations and Future Scope

Reinforcement learning for market reactions: Train an agent on historical news events, sentiment signals, and subsequent market movements to learn how different types of news affect stock prices.

Temporal analysis: Distinguish between breaking news, follow-up coverage, historical references, and repeated articles to avoid treating duplicate information as new signals.

More data sources: Add SEC filings, earnings releases, earnings-call transcripts, analyst reports, macroeconomic data, market news, and additional social-media sources.

Sentiment propagation: The current system propagates sentiment through stock correlations. This can be improved by modeling sector relationships, supply chains, business dependencies, ownership links to estimate how a news event may affect related companies more accurately.

Thank You

- Live Website
- Github Link
- Demo Video

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